

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-inde  
xer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-  
a440f8ab9d8cd6054.jsonl`  
- SHA-256: `573038508aa8670b092273342c7421bfc192bbbe96f4f8d0c54271d8ba1dab9b`  
- Source modified: `2026-06-16T03:25:58+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:24:58.751Z)

Adversarial Claim Verifier (voter 3/3)

Be SKEPTICAL. Try to REFUTE this claim. 2/3 refutations kill it.

Research question

Research question: **What concrete, recent (2022-2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 - ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96-99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval+serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best **recent** methods for (a) turning a dense per-frame signal into well-segmented

actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMN/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k, edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6?16 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6?16 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 3?5 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible.

Claim under review

"ActionFormer is a single-stage, anchor-free temporal action localization model that combines a multiscale feature pyramid with local self-attention (a Transformer) and a lightweight decoder that classifies every moment in time and regresses the corresponding action boundaries, eliminating the need for separate action proposals."

****Source:**** https://dl.acm.org/doi/10.1007/978-3-031-19772-7_29 (primary)

****Supporting quote:**** "The model employs "a multiscale feature representation with local self-attention" and "a light-weighted decoder to classify every moment in time and estimate the corresponding action boundaries." It operates as a single-stage, anchor-free detector without requiring action proposals."

Checklist

1. Is the claim actually supported by the quote, or is it an overreach/misread?
2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

****refuted=true**** if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

****refuted=false**** ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-16T03:25:01.680Z)

I'll verify this claim about ActionFormer. Let me first check the source and search for corroborating/contradicting evidence.

3. assistant (2026-06-16T03:25:05.860Z)

The claim is about a well-known paper (ActionFormer). Let me verify the source URL and the technical claims about its architecture.

4. user (2026-06-16T03:25:08.482Z)

The server returned HTTP 403 Forbidden.

The response body was not retrieved. If this URL requires authentication, use an authenticated tool (e.g. `gh` for GitHub, or an MCP-provided fetch tool) instead of WebFetch.

5. user (2026-06-16T03:25:15.158Z)

Web search results for query: "ActionFormer single-stage anchor-free temporal action localization multiscale Transformer local self-attention"

Links: [{"title": "ActionFormer: Transformer-based Action Localization", "url": "https://www.emergentmind.com/papers/2202.07925"}, {"title": "ActionFormer: Localizing Moments of Actions with Transformers", "url": "https://www.ecva.net/papers/eccv_2022/papers_ECCV/papers/136640485.pdf"}, {"title": "TBT-Former: Learning Temporal Boundary Distributions for Action Localization", "url": "https://www.arxiv.org/pdf/2512.01298"}, {"title": "(PDF) ActionFormer: Localizing Moments of Actions with Transformers", "url": "https://www.researchgate.net/publication/358655346_ActionFormer_Localizing_Moments_of_Actions_with_Transformers"}, {"title": "ActionFormer: Localizing Moments of Actions with Transformers", "url": "https://arxiv.org/pdf/2202.07925"}, {"title": "[PDF] ActionFormer: Localizing Moments of Actions with Transformers | Semantic Scholar", "url": "https://www.semanticscholar.org/paper/ActionFormer:-Localizing-Moments-of-Actions-with-Zhang-Wu/58a3fedc03ab9f5908c077c115eff4c8d2d87660"}, {"title": "ActionFormer: Localizing Moments of Actions with Transformers | Computer Vision ? ECCV 2022", "url": "https://dl.acm.org/doi/10.1007/978-3-031-19772-7_29"}, {"title": "[2512.01298] TBT-Former: Learning Temporal Boundary Distributions for Action Localization", "url": "https://arxiv.org/abs/2512.01298"}]

Based on the search results, I found comprehensive information about ActionFormer:

ActionFormer is a single-stage anchor-free model for temporal action localization. The model combines a multiscale feature representation with local self-attention, and uses a light-weighted decoder to classify every moment in time and estimate the corresponding action boundaries.

ActionFormer is among the first to harness Transformer architectures for single-stage anchor-free temporal action localization, with this strategic adoption of local self-attention enhancing long-range temporal dependencies essential for distinguishing action boundaries. The model incorporates a feature pyramid network (FPN) inspired design to capture actions across varied temporal scales.

ActionFormer achieves impressive results across several major temporal action localization benchmarks including THUMOS14, ActivityNet-1.3, and EPIC-Kitchens 100. Specifically, ActionFormer achieves 71.0% mAP at tIoU = 0.5 on THUMOS14, outperforming the best prior model by 14.1 absolute percentage points.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-16T03:25:22.192Z)

The ACM URL returned 403, but the same paper is confirmed at that DOI in search results. Let me verify the technical claims directly from the authoritative arXiv source.

7. user (2026-06-16T03:25:27.137Z)

ActionFormer Attribute Verification

****Abstract Quote:****

"identify actions in time and recognize their categories in a single shot, without using action proposals or relying on pre-defined anchor windows"

Based on the abstract, here's the confirmation:

1. ****Single-stage****: ? Confirmed - "single shot"
2. ****Anchor-free****: ? Confirmed - "without...relying on pre-defined anchor windows"
3. ****Multiscale feature representation****: ? Confirmed - "multiscale feature representation"
4. ****Local self-attention (Transformer)****: ? Confirmed - "local self-attention"
5. ****Lightweight decoder****: ? Confirmed - "light-weighted decoder"
6. ****Classifies every moment in time****: ? Confirmed - "classify every moment in time"
7. ****Estimates action boundaries****: ? Confirmed - "estimate the corresponding action boundaries"
8. ****Works WITHOUT action proposals****: ? Confirmed - "without using action proposals"

All eight attributes are explicitly described in the abstract using the terminology you specified.

8. assistant (2026-06-16T03:25:33.339Z)

All claims are confirmed directly from the authoritative arXiv abstract. Let me verify one nuance: the claim says "regresses the corresponding action boundaries" ? I should confirm whether ActionFormer regresses boundaries (offsets) versus estimates them, and whether the source DOI genuinely corresponds to this paper.

The arXiv abstract uses "estimate the corresponding action boundaries." The claim's word "regresses" ? ActionFormer regresses distances to action onset/offset at each moment, which is well-documented. The ACM DOI 10.1007/978-3-031-19772-7_29 is confirmed by search results to be the ECCV 2022 ActionFormer paper.

Let me do one adversarial check to ensure there is no qualification or dispute about the "feature pyramid" characterization, since the abstract says "multiscale feature representation" rather than explicitly "feature pyramid."

9. user (2026-06-16T03:25:39.251Z)

ActionFormer Architecture Analysis

Based on the PDF content provided, I can confirm the following about ActionFormer:

****Multiscale Feature Pyramid:**** Yes, ActionFormer employs a multiscale approach. The document references "a multiscale temporal convolutional encoder" as part of its core architecture.

****Boundary Regression:**** Yes, the model regresses action boundaries. The system performs "regression of action boundaries at each time moment" through dedicated regression heads that predict distance offsets to action start and end points.

****Lightweight Decoder:**** The architecture includes "a lightweight convolutional decoder with separate classification and regression heads," which processes features at multiple temporal scales to generate per-moment predictions.

The design enables ActionFormer to detect temporal action localization by combining hierarchical feature extraction with direct boundary offset regression, avoiding the need for dense sliding windows or anchor generation.

[Binary content (application/pdf, 1.7MB) also saved to C:\Users\avidu\claudio\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781580335293-qf7dtp.pdf]

10. assistant (2026-06-16T03:25:43.393Z)

All technical attributes of the claim are confirmed against the authoritative primary sources.
Let me submit my verdict.

11. user (2026-06-16T03:25:53.730Z)

Structured output provided successfully

12. assistant (2026-06-16T03:25:58.273Z)

Verdict submitted: `**refuted = false**` (claim holds, high confidence).

The ActionFormer claim is accurate and well-supported. Every attribute ? single-stage ("single shot"), anchor-free ("without relying on pre-defined anchor windows"), multiscale feature pyramid, local self-attention Transformer, lightweight decoder, per-moment classification, boundary regression, and no action proposals ? is confirmed verbatim against the authoritative primary source (arXiv:2202.07925, the ECCV 2022 paper matching the cited ACM DOI). The only nuance ("estimate" in the abstract vs "regresses" in the claim) is resolved by the paper body, which explicitly uses regression heads predicting onset/offset distance offsets, making the claim technically precise rather than an overreach.